

6.3 Basis



Core skills you'll master in this section:

- ▶ A basis is not unique; a vector can have different coordinates under different bases.
- ▶ Standard basis: the most “natural” unit-square grid.
- ▶ Orthonormal basis: a rotated unit-square grid.
- ▶ Orthogonal basis: basis vectors that are pairwise orthogonal.
- ▶ Non-orthogonal basis: basis vectors that are not orthogonal, but must be linearly independent.
- ▶ Using the inverse matrix to find a vector's coordinates in a new basis.
- ▶ Converting coordinates between different bases.

This section introduces the concept of basis, emphasizing that a basis is not unique, and a vector can have different coordinates under different bases. First, we define the standard basis, which is the most intuitive and natural, like the unit vectors in the Cartesian coordinate system. Then, we discuss the orthonormal basis, whose basis vectors have length 1 and are mutually orthogonal, but need not be the standard basis. Next, we introduce the orthogonal basis, whose vectors are mutually perpendicular but are not required to have length 1. After that, we discuss the non-orthogonal basis, whose basis vectors need not be orthogonal but must be linearly independent. Finally, we explore change of basis, using the RGB and CMY color spaces as examples to illustrate how to transform coordinates between different bases.

A Basis Is Not Unique

$[\mathbf{e}_1, \mathbf{e}_2]$ is just one of infinitely many bases for the plane $\mathbb{R}^2$.

As shown in Figure 1, $[\mathbf{e}_1, \mathbf{e}_2]$, $[\mathbf{v}_1, \mathbf{v}_2]$, and $[\mathbf{w}_1, \mathbf{w}_2]$ are all bases for the plane $\mathbb{R}^2$!

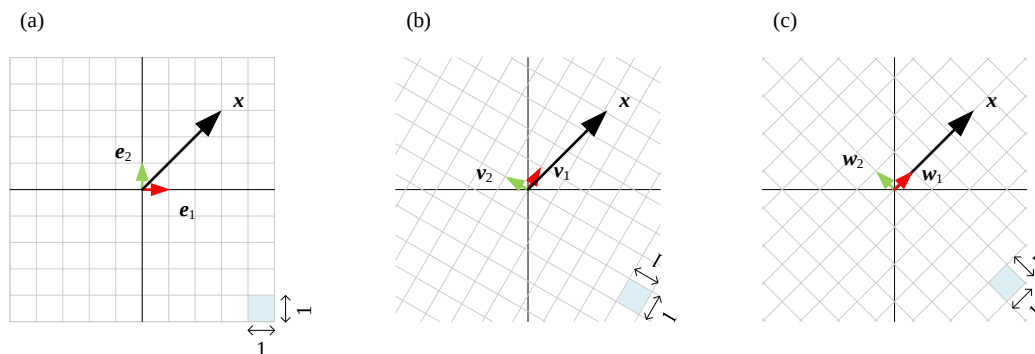


Figure 1. The Position of the Vector $\mathbf{x}$ in Three Different Orthogonal Coordinate Systems

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That is, $\mathbb{R}^2 = \text{span}(\mathbf{e}_1, \mathbf{e}_2) = \text{span}(\mathbf{v}_1, \mathbf{v}_2) = \text{span}(\mathbf{w}_1, \mathbf{w}_2)$.

As shown in Figure 1, on the plane $\mathbb{R}^2$, the vector $\mathbf{x}$ has its own unique coordinates in each of the three bases $[\mathbf{e}_1, \mathbf{e}_2]$, $[\mathbf{v}_1, \mathbf{v}_2]$, and $[\mathbf{w}_1, \mathbf{w}_2]$.

Looking carefully at Figure 1's three different orthogonal rectangular coordinate systems, we find that their grid cells are all unit squares.

You may have already noticed that in Figure 1, $[\mathbf{e}_1, \mathbf{e}_2]$, $[\mathbf{v}_1, \mathbf{v}_2]$, $[\mathbf{w}_1, \mathbf{w}_2]$ — each basis vector is a unit vector, that is, $\|\mathbf{e}_1\| = \|\mathbf{e}_2\| = \|\mathbf{v}_1\| = \|\mathbf{v}_2\| = \|\mathbf{w}_1\| = \|\mathbf{w}_2\| = 1$.

And within each basis, the basis vectors are mutually orthogonal, that is, $\mathbf{e}_1$ is perpendicular to $\mathbf{e}_2$, $\mathbf{v}_1$ is perpendicular to $\mathbf{v}_2$, and $\mathbf{w}_1$ is perpendicular to $\mathbf{w}_2$. In this book, if the basis vectors within a basis are pairwise orthogonal, that basis is called an orthogonal basis. The coordinates of any vector in these orthogonal coordinate systems can be obtained through the orthogonal decomposition of vectors introduced earlier in this book — please review that; this section will also introduce a more convenient method.

If every basis vector in an orthogonal basis has norm 1, that basis is called an orthonormal basis. In Figure 1, the three bases $[\mathbf{e}_1, \mathbf{e}_2]$, $[\mathbf{v}_1, \mathbf{v}_2]$, and $[\mathbf{w}_1, \mathbf{w}_2]$ are all orthonormal bases.

There are infinitely many orthonormal bases that span the plane $\mathbb{R}^2$.

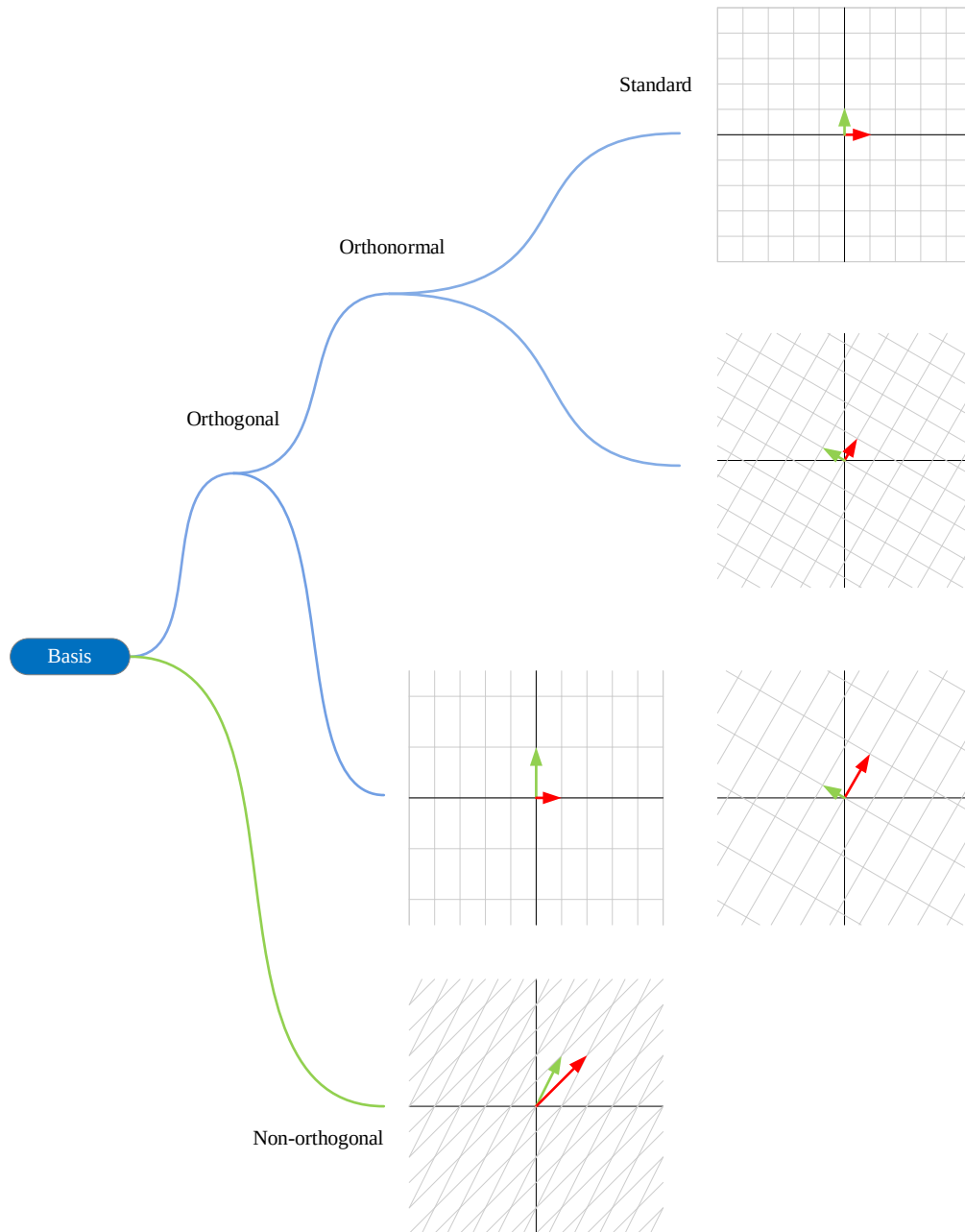


Figure 2. Relationships Among Several Kinds of Bases

$[\mathbf{e}_1, \mathbf{e}_2]$ is clearly a special orthonormal basis; we call it the standard basis (standard basis, natural basis), or simply the standard basis, of the plane $\mathbb{R}^2$.

The word “standard” is given to $[\mathbf{e}_1, \mathbf{e}_2]$ because this basis is the most natural way to represent the plane $\mathbb{R}^2$. $[\mathbf{e}_1, \mathbf{e}_2]$ is also the most common reference frame for the planar rectangular coordinate system. Rotating or mirroring the standard basis gives other orthonormal bases.

But looking them over, only Figure 1 (a)’s grid looks the most natural, which is why $[\mathbf{e}_1, \mathbf{e}_2]$ is called the standard basis.

Clearly, $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$ is the standard basis of $\mathbb{R}^3$, and $[\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_p]$ is the standard basis of $\mathbb{R}^p$.

The “grid cells” that span $\mathbb{R}^2$ need not be unit squares — they can be other squares, rotated squares, rectangles, rotated rectangles, parallelograms, and so on. But generally speaking, these grid cells all share a common feature — they form a grid made of parallelograms that are parallel, equally spaced, and pass through the origin.

Figure 2 summarizes the relationships among several kinds of bases. Next, let's go through each of these bases one by one.

Standard Basis: The Most Natural

Let's first look at the plane $\mathbb{R}^2$'s standard basis $[\mathbf{e}_1, \mathbf{e}_2]$. As shown in Figure 3 (a)

Figure 3's remaining three bases $[-\mathbf{e}_1, \mathbf{e}_2]$, $[\mathbf{e}_1, -\mathbf{e}_2]$, $[-\mathbf{e}_1, -\mathbf{e}_2]$ are also unit-square grids, but they look somehow “off” — not as “natural” as Figure 3 (a)! Only (a) in Figure 3 is the standard basis; the other three bases are orthonormal bases.

The orthonormal basis is a topic this section will introduce later.

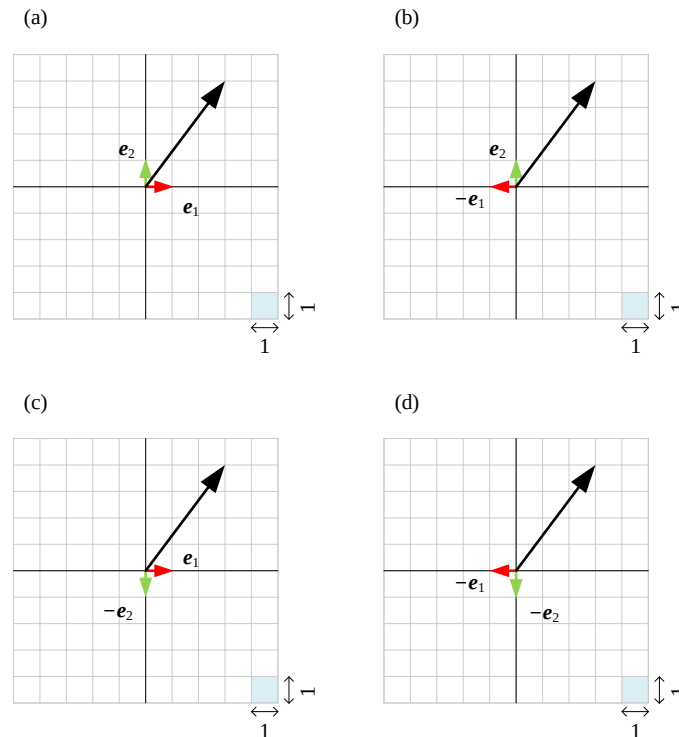


Figure 3. A Few Unit-Square Grids

Clearly, the vector $\mathbf{a} = [3, 4]^T$ has coordinates $(3, 4)$ in $[\mathbf{e}_1, \mathbf{e}_2]$. Let's work this out in detail below.

Any planar vector whose coordinates in the basis $[\mathbf{e}_1, \mathbf{e}_2]$ are (x_1, x_2) can be written as the matrix product

$$x_1\mathbf{e}_1 + x_2\mathbf{e}_2 = [\mathbf{e}_1 \ \mathbf{e}_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The vector $\mathbf{a} = [3, 4]^T$ can be written as

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

To find the coordinates of the vector $\mathbf{a} = [3, 4]^T$ in the basis $[\mathbf{e}_1, \mathbf{e}_2]$, we need to compute

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

where the inverse of the identity matrix is itself.

Next, let's use a similar method to compute the vector $\mathbf{a} = [3, 4]^T$'s coordinates in $[-\mathbf{e}_1, -\mathbf{e}_2]$ as (y_1, y_2) .

The vector $\mathbf{a} = [3, 4]^T$ can be written as

$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

It's easy to compute that

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}^{-1} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -3 \\ -4 \end{bmatrix}$$

This is consistent with what we observed in Figure 3 (d).

Code 1 shows how to compute the coordinates of the vector $\mathbf{a} = [3, 4]^T$ in different bases. Let's go over the key statements below.

- a** The `np.array()` function is used to create a "2D array," that is, the vector $\mathbf{a} = [3, 4]^T$.
- b** Creates several different bases; the commented-out code in the lines that follow provides alternative coordinate systems that are not currently activated.
- c** is the basis that is not commented out; 's square matrix is $\mathbf{A}$, which will be used in the subsequent computation.
- d** Uses `numpy.linalg.inv()` to compute the inverse of the square matrix $\mathbf{A}$.
- e** This line is the goal of our entire code. We want to know: if we switch to a new coordinate system, that is, the new direction represented by the matrix $\mathbf{A}$, then how is the original vector $\mathbf{a} = [3, 4]^T$ expressed in this new system? We use the `@` symbol to denote "matrix multiplication" — that is, we take the inverse matrix $\mathbf{A_inv}$ and multiply it by the original vector $\mathbf{a} = [3, 4]^T$, to get its "coordinates" in the new coordinate system.

Code 1. Computing a Vector's Coordinates in Different Bases |  LA_06_03_01.ipynb

```

## 初始化
import numpy as np

## 定义向量
a = np.array([[3],
              [4]])

## 定义不同基底
# 规范正交基
A = np.array([[-1, 0],
              [0, -1]])

# 规范正交基
# A = np.array([[3/5, -4/5],
#               [4/5, 3/5]])

# 正交基
# A = np.array([[-4/5, -3/2],
#               [3/5, -2]])

# 非正交基
# A = np.array([[2, 1],
#               [1, 3]])

## 计算逆矩阵
A_inv = np.linalg.inv(A)
A_inv

## 计算坐标
A_inv @ a

```



The Standard Basis in RGB Space

As shown in Figure 4, in the RGB color space, the three basis vectors $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ are pairwise orthogonal, so their pairwise inner products are 0:

$$\mathbf{e}_1 \cdot \mathbf{e}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 0, \quad \mathbf{e}_1 \cdot \mathbf{e}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0, \quad \mathbf{e}_2 \cdot \mathbf{e}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0$$

where $\mathbf{e}_1$ represents red, $\mathbf{e}_2$ represents green, and $\mathbf{e}_3$ represents blue.

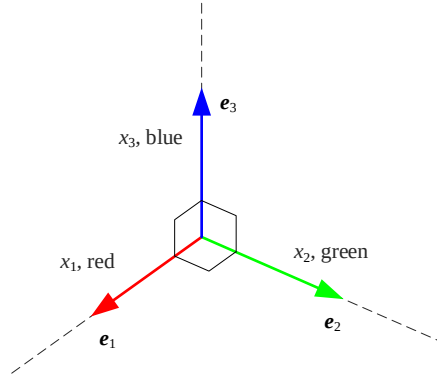


Figure 4. The Three-Primary-Color Space

Moreover, $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ are all unit vectors (direction vectors); their vector length (norm, L2 norm, Euclidean norm) is 1, that is,

$$\|\mathbf{e}_1\|_2 = 1, \quad \|\mathbf{e}_2\|_2 = 1, \quad \|\mathbf{e}_3\|_2 = 1$$

Therefore, in the vector space V of the three-primary-color model, $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$ is V 's standard basis.

Any color can be regarded as a linear combination of the three basis vectors in the standard basis $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$, that is,

$$x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3$$

where x_1 , x_2 , and x_3 all range over $[0, 1]$.

As mentioned repeatedly earlier, to be precise, the RGB three-primary-color space is not a true vector space as described earlier in this book, because x_1 , x_2 , and x_3 are subject to a range restriction. A true vector space has no such restriction. Except for the zero vector $\mathbf{0}$, a genuine vector space extends infinitely.

Any color vector whose coordinates in the basis $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$ are (x_1, x_2, x_3) can be written as the matrix product

$$x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3 = [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

For example, the white vector can be written as

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

To find the white vector's coordinates in the basis $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$, we only need to carry out the following computation

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

You should have noticed that we've repeatedly emphasized "orthogonal," that is, perpendicular. When Descartes first proposed the concept of "coordinates," he did not specify that they had to be

“rectangular coordinates.” In fact, he believed that any two non-parallel lines could be used to describe coordinates in a two-dimensional plane. We favor the rectangular coordinate system because it really is convenient and natural.

Symmetric — the Pythagorean theorem in geometry relies on right angles. The rectangular coordinate system makes vector operations intuitive; for example, computing a vector's length, angle, and inner product can all directly apply the Pythagorean theorem and trigonometric functions.

In addition, the orthogonal basis in a rectangular coordinate system can greatly simplify coordinate transformations, making operations such as projection, rotation, and scaling more efficient. Rectangular coordinate systems are widely used in engineering, physics, computer science, and many other fields, because they not only match people's intuition but also reduce computational complexity.

However, although the rectangular coordinate system is the most commonly used, it is not the only choice. In certain application scenarios, such as fluid dynamics, image transformation, or computer graphics, using a non-orthogonal basis may be better suited to a specific problem. For example, an oblique coordinate system can more naturally describe certain physical phenomena, while a polar coordinate system can more concisely express circular motion. Different coordinate systems each have their own advantages; the key is choosing the representation best suited to the problem at hand.

Orthonormal Basis: Orthogonal Unit Vectors

Simply put, an orthonormal basis is a basis made up of pairwise orthogonal unit vectors. It's not required to be as “natural” as the standard basis.

Figure 5 shows four orthonormal bases.

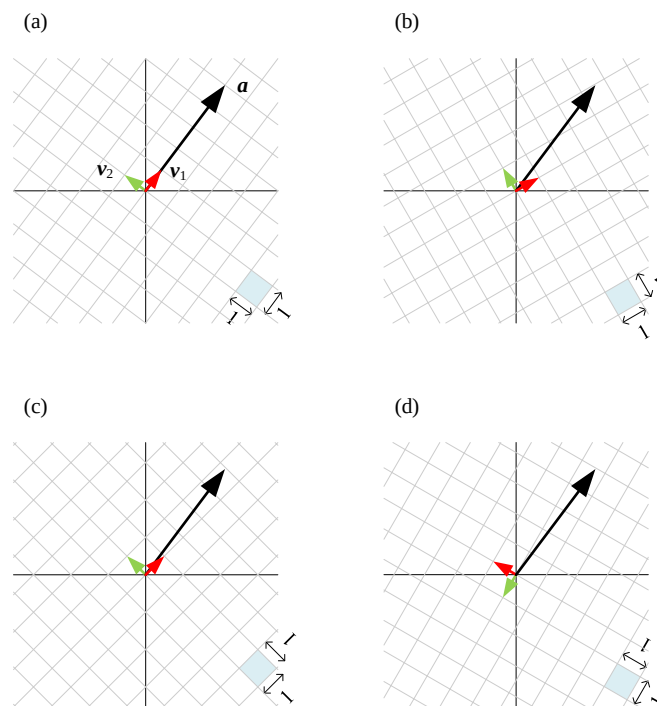


Figure 5. A Few Rotated Unit-Square Grids (Orthonormal Bases)

Taking Figure 5 (a) as an example, this orthonormal basis is

$$\mathbf{V} = [\mathbf{v}_1 \quad \mathbf{v}_2] = \begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}$$

$\mathbf{V}$ and its transpose $\mathbf{V}^T$, the product is

$$\begin{aligned} \mathbf{V}\mathbf{V}^T &= [\mathbf{v}_1 \quad \mathbf{v}_2] \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \end{bmatrix} = \mathbf{v}_1 @ \mathbf{v}_1^T + \mathbf{v}_2 @ \mathbf{v}_2^T \\ &= \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} @ \begin{bmatrix} 3/5 & 4/5 \end{bmatrix}^T + \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix} @ \begin{bmatrix} -4/5 & 3/5 \end{bmatrix}^T \\ &= \begin{bmatrix} 9/25 & 12/25 \\ 12/25 & 16/25 \end{bmatrix} + \begin{bmatrix} 16/25 & -12/25 \\ -12/25 & 9/25 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I} \end{aligned}$$

Conversely, $\mathbf{V}^T\mathbf{V}$ is also an identity matrix of the same shape, that is, $\mathbf{V}\mathbf{V}^T = \mathbf{V}^T\mathbf{V} = \mathbf{I}$, please compute this yourself. This shows that $\mathbf{V}$ is an orthogonal matrix.

Let's compute the vector $\mathbf{a} = [3, 4]^T$'s coordinates in Figure 5 (a) as (y_1, y_2) .

The vector $\mathbf{a} = [3, 4]^T$ can be written as

$$\begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

It's easy to compute that

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}^{-1} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3/5 & 4/5 \\ -4/5 & 3/5 \end{bmatrix}^T \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

That is, the vector $\mathbf{a}$'s coordinates in this orthonormal basis are (5, 0), which is consistent with what we observed in Figure 5 (a).



LA_06_03_02.ipynb plots orthonormal basis grids; please study it on your own.

Orthogonal Basis: Satisfying Only Orthogonality

An orthonormal basis requires that the basis vectors all be unit vectors and be pairwise orthogonal. Dropping the "unit vector" requirement while keeping only the "orthogonal" constraint gives an orthogonal basis.

As shown in Figure 6, an orthogonal basis's grid can be a square, a rotated square, a rectangle, or a rotated rectangle.

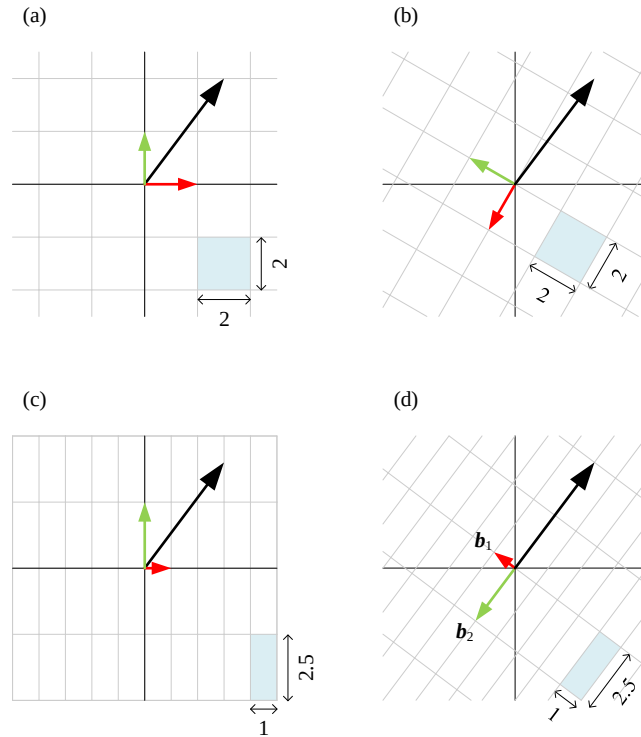


Figure 6. A Few Orthogonal Bases

Taking Figure 6 (d) as an example, the two basis vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ are

$$\mathbf{b}_1 = \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -3/2 \\ -2 \end{bmatrix}$$

$\mathbf{b}_1$ and $\mathbf{b}_2$'s inner product is 0, so the two are orthogonal; but $\mathbf{b}_2$ is not a unit vector.

Let's compute the vector $\mathbf{a} = [3, 4]^T$'s coordinates in Figure 6 (d) as (y_1, y_2) .

The vector $\mathbf{a} = [3, 4]^T$ can be written as

$$\begin{bmatrix} -4/5 & -3/2 \\ 3/5 & -2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

It's easy to compute that

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} -4/5 & -3/2 \\ 3/5 & -2 \end{bmatrix}^{-1} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -0.8 & 0.6 \\ -0.24 & -0.32 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

That is, the vector $\mathbf{a}$'s coordinates in this orthogonal basis are $(0, -2)$, which is consistent with what we observed in Figure 6 (d).

Non-Orthogonal Basis: Not Orthogonal

An orthogonal basis requires the basis vectors to be pairwise orthogonal; relaxing the “pairwise orthogonal” constraint gives a non-orthogonal basis. Note that the basis vectors of a non-orthogonal basis are still required to be linearly independent — this is a condition basis vectors must always satisfy.

Figure 7 shows a few non-orthogonal bases.

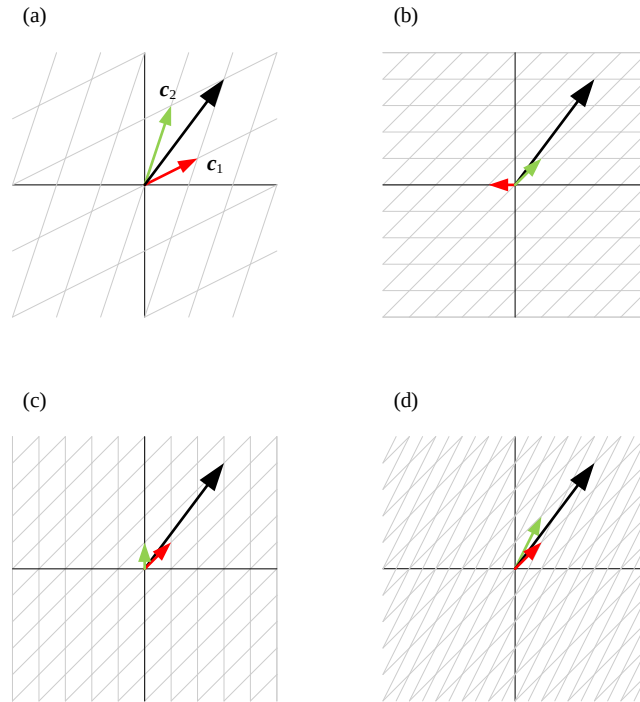


Figure 7. A Few Non-Orthogonal Bases

Taking Figure 7 (a) as an example, the two basis vectors $\mathbf{c}_1$ and $\mathbf{c}_2$ are

$$\mathbf{c}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$\mathbf{c}_1$ and $\mathbf{c}_2$'s inner product is nonzero, so the two are non-orthogonal.

Let's compute the vector $\mathbf{a} = [3, 4]^T$'s coordinates in Figure 7 (a) as (y_1, y_2) .

The vector $\mathbf{a} = [3, 4]^T$ can be written as

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

It's easy to compute that

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}^{-1} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.2 \\ -0.2 & 0.4 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

That is, the vector $\mathbf{a}$'s coordinates in this non-orthogonal basis are $(1, 1)$, which is consistent with what we observed in Figure 7 (a).

A Non-Orthogonal Basis in RGB Space

$\mathbf{e}_1$ ($[1, 0, 0]^T$ red), $\mathbf{e}_2$ ($[0, 1, 0]^T$ green), and $\mathbf{e}_3$ ($[0, 0, 1]^T$ blue) — combining any two of these three basis vectors constructs three vectors: $\mathbf{a}_1$ ($[0, 1, 1]^T$ cyan), $\mathbf{a}_2$ ($[1, 0, 1]^T$ magenta), and $\mathbf{a}_3$ ($[1, 1, 0]^T$ yellow):

$$\mathbf{a}_1 = \mathbf{e}_2 + \mathbf{e}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{a}_2 = \mathbf{e}_1 + \mathbf{e}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{a}_3 = \mathbf{e}_1 + \mathbf{e}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

As shown in Figure 8, $\mathbf{a}_1$ is equivalent to a linear combination of $\mathbf{e}_2$ and $\mathbf{e}_3$, $\mathbf{a}_2$ is equivalent to a linear combination of $\mathbf{e}_1$ and $\mathbf{e}_3$, and $\mathbf{a}_3$ is equivalent to a linear combination of $\mathbf{e}_1$ and $\mathbf{e}_2$.

$\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ are linearly independent, so $[\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3]$ can also be a basis for constructing the three-dimensional color space!

The four-color printing model (CMYK color model) is based on the basis $[\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3]$. The four letters in CMYK stand for cyan, magenta, yellow, and black. In this section, we only consider three colors: cyan, magenta, and yellow.

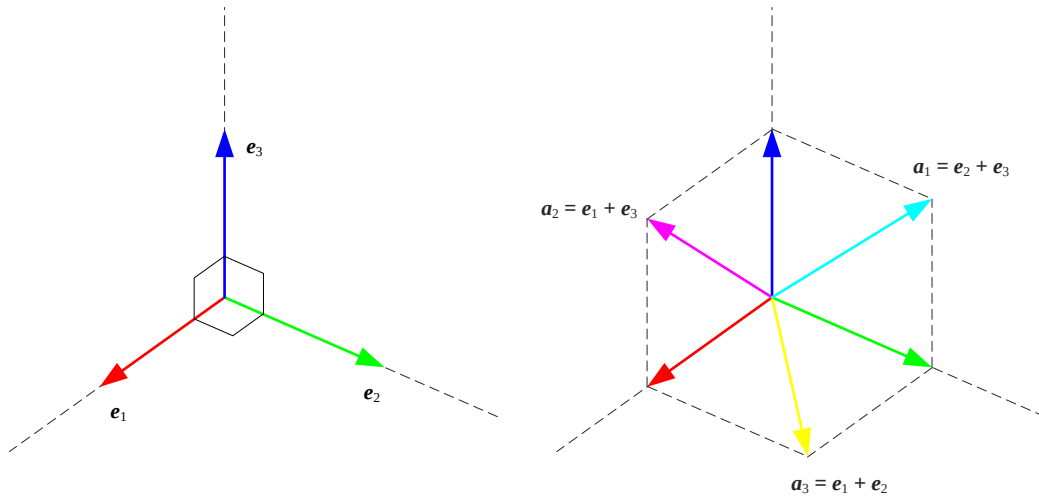


Figure 8. From an Orthogonal Basis to a Non-Orthogonal Basis

$\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ are not pairwise orthogonal. Computation shows that $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ have a pairwise angle of 60°

$$\begin{aligned} \cos \theta_{\mathbf{a}_1, \mathbf{a}_2} &= \frac{\mathbf{a}_1 \cdot \mathbf{a}_2}{\|\mathbf{a}_1\| \|\mathbf{a}_2\|} = \frac{1}{\sqrt{2} \times \sqrt{2}} = \frac{1}{2} \\ \cos \theta_{\mathbf{a}_1, \mathbf{a}_3} &= \frac{\mathbf{a}_1 \cdot \mathbf{a}_3}{\|\mathbf{a}_1\| \|\mathbf{a}_3\|} = \frac{1}{\sqrt{2} \times \sqrt{2}} = \frac{1}{2} \\ \cos \theta_{\mathbf{a}_2, \mathbf{a}_3} &= \frac{\mathbf{a}_2 \cdot \mathbf{a}_3}{\|\mathbf{a}_2\| \|\mathbf{a}_3\|} = \frac{1}{\sqrt{2} \times \sqrt{2}} = \frac{1}{2} \end{aligned}$$

That is, $[\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3]$ is a non-orthogonal basis.

The white-light vector's coordinates in $[\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3]$ are (y_1, y_2, y_3) , satisfying the following matrix product

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

To find the white vector's coordinates in the basis $[a_1, a_2, a_3]$, we only need to carry out the following computation

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.5 & 0.5 & 0.5 \\ 0.5 & -0.5 & 0.5 \\ 0.5 & 0.5 & -0.5 \end{bmatrix} @ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

There are infinitely many non-orthogonal bases in RGB space. Given any three color vectors, as long as they are not coplanar (that is, linearly independent), they can span the entire RGB color space.

Change of Basis

Simply put, a change of basis is the transformation of the same vector's coordinates between different bases.

Suppose a p -dimensional vector space V has two bases, $U = [u_1, u_2, \dots, u_p]$ and $W = [w_1, w_2, \dots, w_p]$. This means that U and W are both invertible.

For any vector v in the space V , its coordinates in the two bases U and W are x and y , respectively, that is,

$$v = x_1 u_1 + x_2 u_2 + \dots + x_p u_p = \begin{bmatrix} u_1 & u_2 & \dots & u_p \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} = Ux$$

$$v = y_1 w_1 + y_2 w_2 + \dots + y_p w_p = \begin{bmatrix} w_1 & w_2 & \dots & w_p \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{bmatrix} = Wy$$

And Ux and Wy are both the coordinates of the vector v in the standard basis, so clearly the two are equal. Combining the two equations above, we get

$$Ux = Wy$$

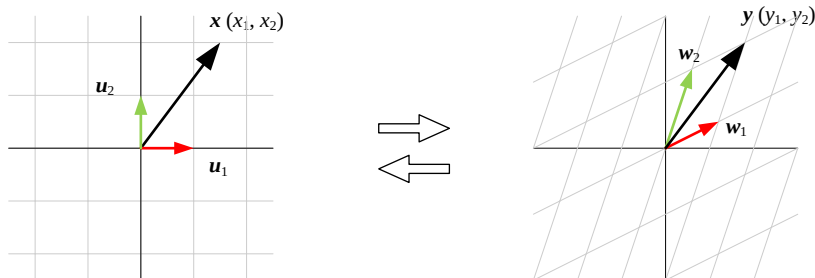


Figure 9. Change of Basis

If we know $\mathbf{v}$'s coordinates $\mathbf{U}$ in the basis $\mathbf{x}$, and want to find its coordinates $\mathbf{W}$ in $\mathbf{y}$, we can use the equation below

$$\mathbf{y} = \mathbf{W}^{-1}\mathbf{U}\mathbf{x}$$

Conversely, if we know $\mathbf{v}$'s coordinates $\mathbf{W}$ in the basis $\mathbf{y}$, and want to find its coordinates $\mathbf{U}$ in $\mathbf{x}$, we can use the equation below

$$\mathbf{x} = \mathbf{U}^{-1}\mathbf{W}\mathbf{y}$$



LA_06_03_03.ipynb shows how to perform a change of basis in code; please study it on your own.

RGB $\leftrightarrow$ CMY

Below, let's again use a color chart as an example to explain change of basis.

In the RGB color chart, $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$ is the standard basis of the color space. In the CMY color chart, $[\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3]$ is a non-orthogonal basis of the color space. We can use a change of basis to convert from RGB mode to CMY mode.

The figure below shows the conversion relationship between the basis $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$ and the basis $[\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3]$.

⚠ Note that in the printing industry, the actual conversion between RGB and CMYK is much more complex than the conversion above.

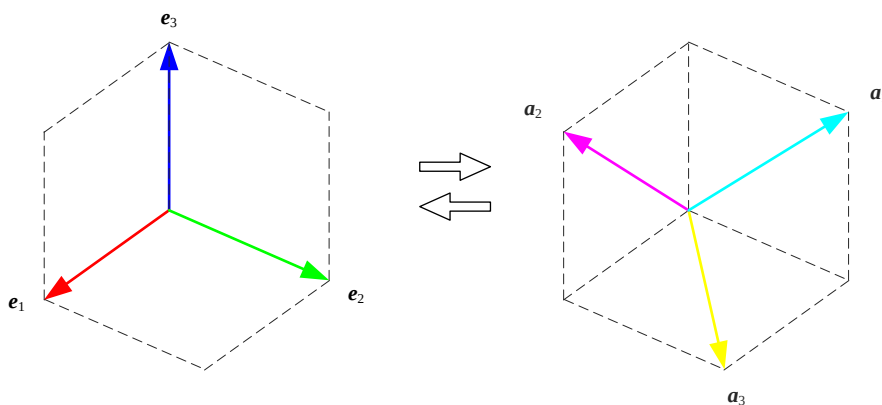


Figure 10. Converting Between the Basis $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$ and the Basis $[\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3]$

"Pure red"'s coordinates in the basis $[\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3]$ $\mathbf{y}$ can be found by solving the following system of linear equations:

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \mathbf{I}_{3 \times 3} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Then compute

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} -0.5 & 0.5 & 0.5 \\ 0.5 & -0.5 & 0.5 \\ 0.5 & 0.5 & -0.5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

giving “pure red”'s coordinates in the basis $[\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3]$ as $(-0.5, 0.5, 0.5)$.

 Please compute on your own the coordinates of green, blue, white, cyan, magenta, and yellow in the basis $[\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3]$.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please determine which of the following vector groups can serve as a basis for the two-dimensional plane.

▶ $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix}$

▶ $\begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

▶ $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

▶ $\begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$

Q2. Please determine which of the following are orthogonal bases for the two-dimensional plane.

▶ $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix}$

▶ $\begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 4 \\ -3 \end{bmatrix}$

▶ $\begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

Q3. Please determine which of the following vector groups can serve as a basis for three-dimensional space.

▶ $\begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

▶ $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$

▶ $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

Q4. What are the vector $\mathbf{a} = [3, 4]^T$'s coordinates in each of the following different bases?

▶ $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix}$

▶ $\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

▶ $\begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$

Q5. Compute the coordinates of red, green, and blue in CMY.

